

Spectral Clustering vs K-Means, GMM and Agglomerative Clustering: An Analytical Report

Clustering continues to play a central role in modern data science and machine learning, driven by its ability to uncover hidden groupings within datasets without the need for labeled outcomes. Unlike supervised learning, clustering does not rely on predefined outputs but instead uses inherent patterns in the data to assign observations to different groups. This makes it extremely valuable across many disciplines, including medical diagnostics, business intelligence, genomics, social network analysis, and visual recognition. The report presented here is an analytical exploration of four widely known clustering algorithms: K-Means, Gaussian Mixture Models (GMM), Agglomerative Clustering, and Spectral Clustering with the objective of evaluating how their behaviour differs when confronted with nonlinear datasets. This work demonstrates why spectral clustering, which leverages graph theory and eigenvalues, outperforms traditional methods when dealing with irregular or curved distributions.

The interest in clustering arises from a simple intuition: items in the same group should share similarities, whereas items belonging to other groups should be more different. Though easy to state conceptually, this idea becomes challenging when operationalized, because it depends heavily on how similarity is defined and measured. Algorithms such as K-Means rely on Euclidean distance, which assumes that clusters are spherical and centrally aligned. This assumption frequently fails in complex real-world datasets where groups may be elongated, intertwined or separated by nonlinear boundaries. Furthermore, real datasets rarely tell us how many clusters exist, which further complicates algorithm selection. These limitations highlight an important reality: although clustering is one of the oldest unsupervised learning approaches, it is not universally applicable in the same form across all contexts.

To make this point clear, our study sought to compare clustering methods using the well-known “Two-Moons” dataset. The dataset consists of two curved crescent-shaped regions that follow a non-linear pattern. This dataset was selected intentionally because it exposes the weaknesses of distance-based clustering and represents a classical challenge in unsupervised learning research. When the dataset is visualized, one can clearly see two distinct shapes, but because the shapes are curved and partially intertwined, algorithms that measure straight-line distances between points will struggle to separate them correctly. The aim of the experiment was not simply to demonstrate that Spectral Clustering performs well, but to allow a practical and quantitative comparison across algorithms as their internal mechanisms interact differently with non-linear data structures.

Before analysing the performance outcomes, it is essential to review how each of the four algorithms works. K-Means remains perhaps the most famous clustering method, largely due to its speed, simplicity, and interpretability. It is widely applied across industry and academic contexts because it works effectively on large datasets and performs well when clusters are spherical. Despite this popularity, K-Means has significant weaknesses: it requires users to pre-define the number of clusters, it is sensitive to initial centroid placement, and most importantly, it assumes that clusters are compact and equidistant, something that is not always true. Gaussian Mixture Modelling (GMM) offers a probabilistic alternative to K-Means, with the flexibility to model a dataset as a mixture of overlapping distributions. However, GMM still assumes that clusters are Gaussian in shape, and performance decreases when clusters have non-normal geometric properties.

Agglomerative Clustering approaches the problem from a hierarchical perspective rather than a distance-centric one. Instead of dividing data into pre-defined partitions, it begins by treating every data point as its own cluster and then repeatedly merges the most similar clusters based on a linkage rule. The method is particularly appealing for datasets where the structure is nested or non-linear. Nevertheless, the approach has drawbacks: it is computationally expensive at scale, and because it uses a greedy procedure, merging

closest clusters at each step, it cannot reverse past decisions even when later evidence suggests that an alternative merger would have been more optimal.

The most conceptually different method among the four is Spectral Clustering. Rather than applying distance-based clustering directly in the original space, it uses graph theory to represent the dataset. A similarity graph is constructed by calculating how strongly each pair of points is connected, for example by checking whether they are close neighbors. From this graph, a Laplacian matrix is constructed, and eigenvalue decomposition is applied. This transformation projects the dataset into a new low-dimensional space where clusters become linearly separable. Only then is a standard clustering algorithm often K-Means — applied in the transformed space. This two-step process, though more mathematically complex, enables Spectral Clustering to succeed where classical clustering fails because it respects connectivity rather than spatial separation alone.

To evaluate the behaviour of these algorithms, we tested them on the Two-Moons dataset under the same experimental conditions. The results were stark. K-Means incorrectly split the dataset by drawing a straight-line decision boundary and assigning large portions of both moons to the same cluster. The result was a visual and mathematically flawed representation of the real structure of the data. GMM responded slightly better, capturing some of the curvature of the moons, but it still failed to appreciate the non-linear nature of the distribution, leading to significant misclassification. In contrast, both Agglomerative Clustering and Spectral Clustering produced cluster assignments that preserved the two curved moons almost perfectly. Spectral Clustering achieved an Adjusted Rand Index (ARI) of 1.00, which corresponds to a perfect match with the true underlying groupings, whereas the ARI for K-Means and GMM dropped significantly. A direct comparison of cluster assignments revealed that 24.83% of points differed between Spectral Clustering and K-Means, 15.17% differed between Spectral Clustering and GMM, and 0% differed between Spectral Clustering and Agglomerative Clustering.

These findings demonstrate an important nuance: clustering success is not merely about producing compact clusters. In this experiment, K-Means generated a higher silhouette score than Spectral Clustering, yet its grouping was wrong in the context of the dataset's intrinsic pattern. This breakdown indicates that a high silhouette score can provide a false sense of confidence if the dataset violates an algorithm's mathematical assumptions. Spectral Clustering performed better not because it formed tighter clusters, but because it interpreted connectivity, a characteristic that cannot be captured by Euclidean distance alone. In real-world applications where data rarely forms perfect spheres, this distinction becomes critical.

Another noteworthy insight from the experiment relates to the interpretability of clustering outcomes. When clusters are visually represented on the Two-Moons dataset, even a human observer can intuitively spot that there are two clear groups. Yet algorithms like K-Means, which are among the most frequently applied in industry, fail to produce the intuitive result. This misalignment between algorithmic output and human intuition highlights a broader lesson in data science — algorithms are not universally correct simply because they are popular or computationally efficient. Understanding the mathematical assumptions behind each algorithm is critical to matching it with the structure of the dataset. In practical applications, analysts and machine learning engineers should never rely on a “default” clustering algorithm without first examining whether the data geometry supports it.

The strengths and weaknesses of the algorithms tested carry strong implications for real-world applications. For example, clustering in healthcare might involve grouping patients based on symptoms, genetic sequences or medical imaging. Such data rarely forms neat circles or spherical patterns. Clustering in finance might involve grouping clients by behavioural patterns that change sharply across time, producing irregular clusters in time-series space. Clustering in ecology might involve mapping animal movement paths that form curved trajectories influenced by terrain and environmental constraints. In all these cases, methods like Spectral Clustering that incorporate connectivity, neighborhood relationships and non-linear

geometry tend to outperform distance-based approaches, particularly when the data does not naturally cluster around centroids.

Yet even the strong performance of Spectral Clustering does not imply that it is universally the best algorithm. While it excels on small to medium-scale nonlinear datasets, its computational requirements grow rapidly with dataset size because constructing a similarity graph and computing eigenvalues is expensive. In environments where datasets scale into millions of rows, Spectral Clustering becomes impractical without approximation techniques or high-performance computing. Agglomerative Clustering, though equally effective on the Two-Moons dataset, also suffers from scalability challenges due to its hierarchical nature. In contrast, K-Means continues to dominate industry contexts not because it is always best but because its simplicity and speed make it ideal when data is spherical, high-dimensional and massive in scale.

Therefore, the deeper lesson from this project is that clustering success depends not on the popularity of the method but on the match between algorithm and data behaviour. A competent data scientist must first identify whether the data structure is linear or non-linear, whether clusters are expected to be compact or elongated, whether cluster density is uniform or uneven, and whether relationships are based on absolute distance or neighborhood connectivity. With this knowledge, algorithm selection becomes a strategic decision rather than a guess. When a dataset contains curved or layered structures, Spectral Clustering provides a powerful geometric lens for separating groups that would appear inseparable if analysed using only straight-line distances.

The evaluation also exposed an important practical consideration: evaluation metrics for clustering are not neutral. The silhouette coefficient, for instance, measures compactness and separation, but ignores the actual semantic meaning of clusters. This means that a clustering solution may score highly despite misrepresenting the real grouping of the dataset. The Adjusted Rand Index (ARI), on the other hand, directly measures correspondence between true labels and predicted labels when ground truth is available. Our experiment demonstrates the importance of triangulating evaluation metrics — no single metric can capture all dimensions of clustering quality. The combination of ARI, visual inspection and domain knowledge provides a more rounded basis for deciding whether clustering outcomes are meaningful.

It is also useful to consider the educational perspective of this experiment. The Two-Moons dataset is widely used in machine learning courses not because it resembles real commercial datasets but because it teaches students to recognise the limitations of assumptions. Algorithms that work beautifully on synthetic spherical datasets fail when geometry changes. This realisation is foundational for the development of analytical maturity — the ability to recognise that data shapes dictate the analytical approach, and that even mathematically elegant algorithms can break under the wrong conditions. Students and practitioners alike must take responsibility for validating whether a given model aligns with the behaviour of the dataset rather than assuming that a frequently used method will automatically produce reliable results.

From a methodological standpoint, the experiment followed a structured process. The Two-Moons dataset was generated with noise to introduce realistic imperfections, representing the uncertainty and variability found in real data. Each of the four clustering algorithms was applied under equivalent conditions to ensure fairness of comparison. Visualizations were used to plot the resulting clusters and highlight differences in grouping among the algorithms. A comparison table was developed to compute the proportion of points that differed between cluster assignments. Finally, Adjusted Rand Index and silhouette scores were analysed to provide quantitative assessments of correctness and compactness.

On a broader level, the experiment reinforces the importance of matching machine learning tools to the right problem context. Clustering remains a fundamental step in many analytics pipelines, especially when labels are absent or unknown. However, the experiment demonstrates that the geometry and feature relationships of the dataset must be considered before selecting the algorithm. If cluster boundaries are curved, layered or irregular, Spectral Clustering becomes an appropriate choice because it uses eigenvalue

decomposition to identify low-dimensional manifolds that naturally separate clusters. When datasets are Gaussian in nature, GMM becomes extremely powerful because it models probability distributions rather than discrete membership. When dataset size is extremely large and clusters are circular, K-Means dominates in performance and scalability. When nested relationships and hierarchies matter, Agglomerative Clustering becomes the most insightful algorithm. Thus, success in clustering depends on informed selection rather than blind application.

Conclusion

The comparative experiment using the Two-Moons dataset has demonstrated that clustering algorithms are not interchangeable and that their effectiveness depends fundamentally on the geometry and structure of the data. K-Means and Gaussian Mixture Models, though widely used, produced flawed results on the non-linear dataset because their mathematical assumptions favor spherical clusters and straight-line separation. In contrast, Agglomerative Clustering and Spectral Clustering, particularly the latter provided an accurate representation of the natural structure of the dataset. Spectral Clustering achieved perfect classification ($ARI = 1.00$) by leveraging similarity graphs and eigenvalue decomposition to capture non-linear connectivity.

Therefore, the central conclusion of this project is that clustering accuracy does not depend on computational popularity but on selecting an algorithm whose assumptions match the behaviour of the dataset. Understanding data geometry is the most important prerequisite to effective clustering.

References

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Appendix — Code and Methodology